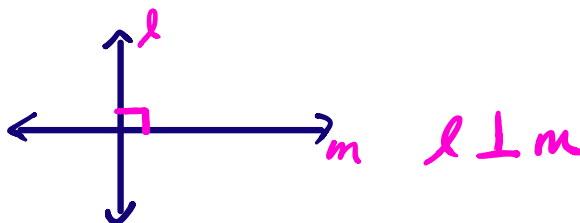


Vocab and More

Perpendicular –



2 lines intersect and form right angles

p.84 **Postulate (or axiom)** - (unproven)
a statement accepted to be TRUE

Theorem - (Vertical Angle Theorem)
a proven statement that is TRUE

Segment Addition Postulate -



$AB + BC = AC$

Algebraic

Properties of Equality – p. 92

1. **Addition Property of Equality** -

If $a = b$, then $a + c = b + c$ (add c)

2. **Subtraction Property of Equality** -

(subtract c)

3. **Multiplication Property of Equality** -

(multiply c)

4. **Division Property of Equality** -

($\div c$)

5. **Substitution Property of Equality** -

⑥ Reflexive Property ^{of =} – p. 94

$$AB = BA$$

$$m\angle A = m\angle A$$

$$AB = AB$$

7. Symmetric Property ^{of =}

$$S = X$$

$$X = S \text{ Symm. Prop.} =$$

If $AB = CD$, then $CD = AB$.

⑧ Transitive Property ^{of =}

If $a = b$ and $b = c$, then $a = c$.

If $a = b$ and $a = c$, then $b = c$.

Other Properties – p. 93

Distributive Property -

Example: Solve the Equation. Justify each step.

$$4 = -10x + 6(2 - x)$$

Given

$$4 = -10x + 12 - 6x$$

Distributive Prop

$$4 = -16x + 12$$

Simplify (combine like terms)

$$-8 = -16x$$

Subtraction Prop (-12)

$$\frac{1}{2} = x$$

Division Prop ($\div -16$)

$$x = \frac{1}{2}$$

Symmetric Prop =

More 2-4 Notes 2-Column Proof How to set it up

Required

Picture, Given info, What to prove

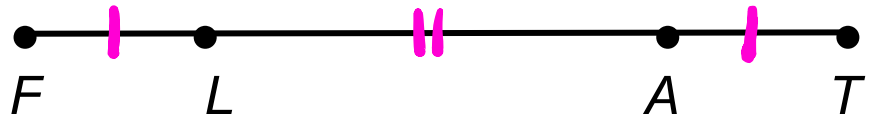
<u>Statements</u>	<u>Reasons</u>
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1.
2.

1. Givens
2. Definitions
Postulates
Theorems
Properties

Practice Proof:

Given: $FL = AT$



Prove: $FA = LT$

<u>Statements</u>	<u>Reasons</u>
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1. $FL = AT$

1. given

2. $LA = LA$

2. Reflexive Prop. =

3. $FL + LA = AT + LA$

3. Addition Prop = (+LA)

4. $FL + LA = FA$;
 $LA + AT = LT$

4. Segment Addition Post.

5. $FA = LT$

5. Substitution Property of =